



INDUCTION COIL

Presentation

By Soumya D. Das, 9A

01

Inventor

Invented by Nikola Tesla in 1891 for wireless electricity.

02

Voltage Output

Produces high-voltage, low-current electrical discharges.

03

Resonance

Uses resonant circuits to generate powerful electrical arcs.

04

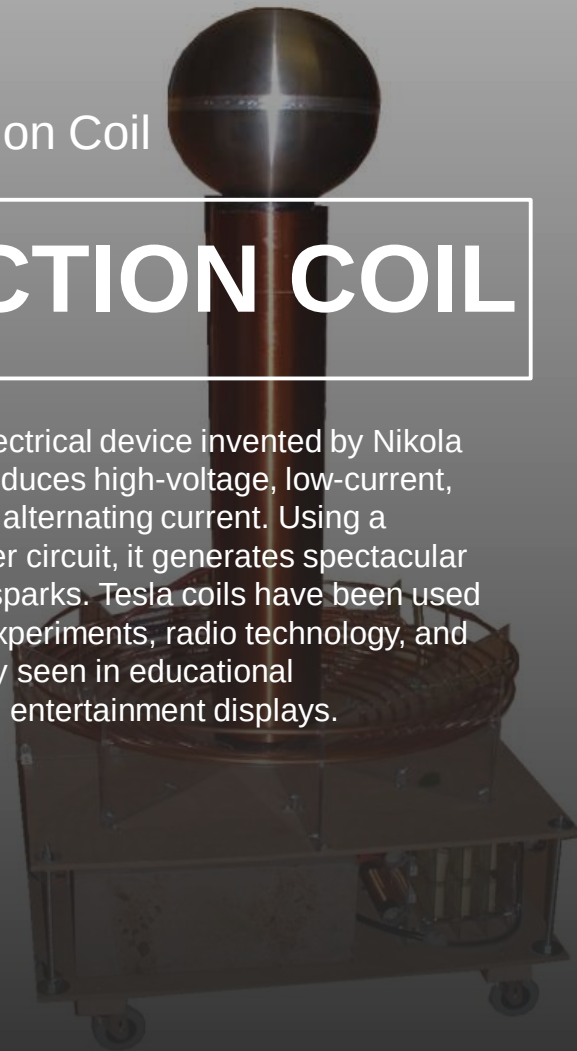
Modern Use

Commonly used in demonstrations and scientific experiments today.

About Induction Coil

INDUCTION COIL

A Tesla coil is an electrical device invented by Nikola Tesla in 1891. It produces high-voltage, low-current, and high-frequency alternating current. Using a resonant transformer circuit, it generates spectacular electrical arcs and sparks. Tesla coils have been used in wireless power experiments, radio technology, and today are commonly seen in educational demonstrations and entertainment displays.





Wireless electricity transmission transfers power without wires, using electromagnetic fields. Pioneered by Nikola Tesla, it enables power delivery over short distances, like wireless charging. Ongoing research aims for broader applications.



Electric arcs in a Tesla coil are bright, high-voltage discharges, demonstrating the coil's ability to produce powerful electrical sparks.

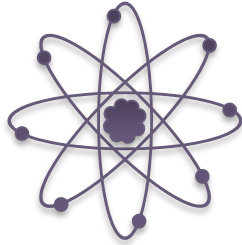


BENEFITS OF INDUCTION COIL



EDUCATIONAL

Provides hands-on experience with electrical phenomena, enhancing understanding of high-voltage and resonance principles in a visually engaging way.



SCIENTIFIC

Aids research by demonstrating electromagnetic fields, resonant circuits, and high-voltage effects, contributing to experimental knowledge and technology development.



WIRELESS

Demonstrates principles of wireless power transmission and resonant energy transfer, showcasing potential for future technology applications.



RESEARCH

Inspires advancements in electrical engineering and communication technologies, contributing to ongoing innovations in the field.

INDUCTION COIL USES

High Voltage Testing

Used for testing electrical insulation and components by simulating high-voltage conditions in a controlled environment.



Medical Equipment

Employed in some medical devices, such as high-frequency diathermy machines, for therapeutic heating and treatment.



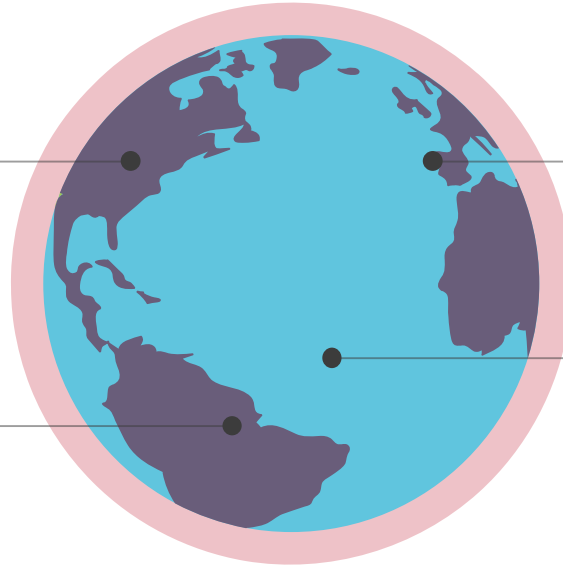
Wireless Charging

Powers devices like smartphones and electric toothbrushes through resonant inductive coupling, demonstrating practical wireless energy transfer.

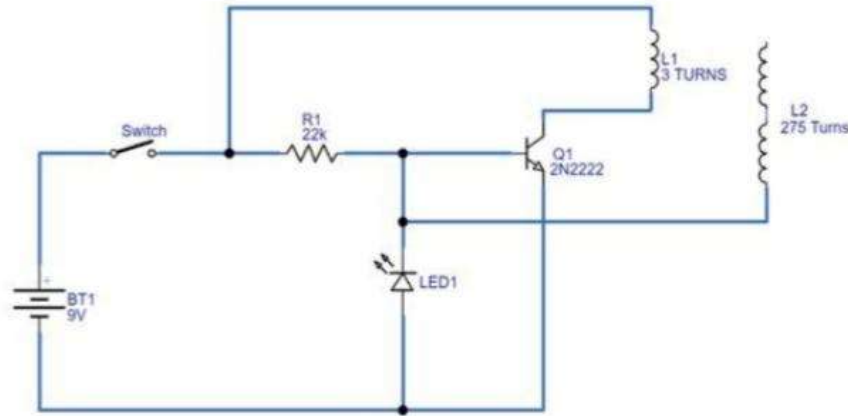


Special Effects

Creates dramatic visual effects in films, theater productions, and live performances, enhancing visual appeal with electrical displays.



CIRCUIT WORKING AND DIAGRAM OF INDUCTION COIL

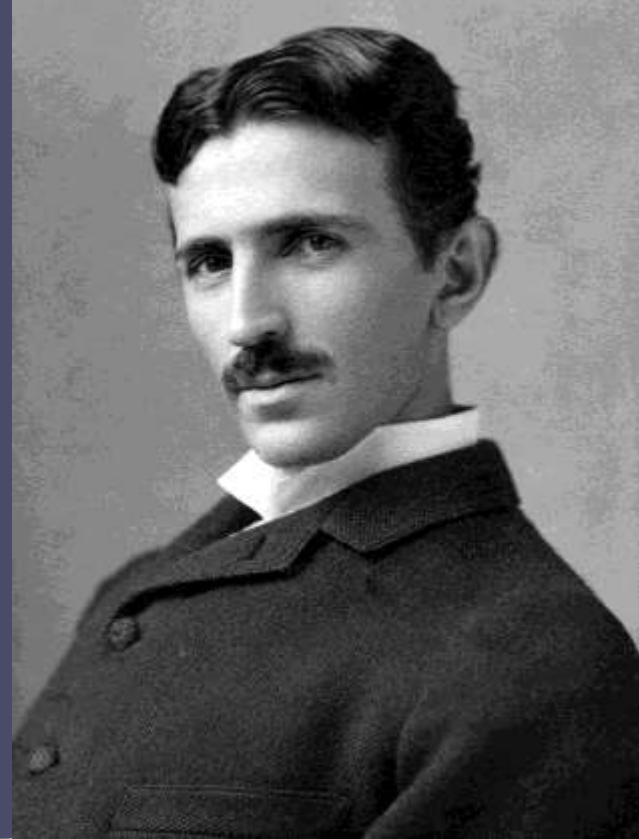


The **9V battery (BT1)** provides the primary power, and the **switch** controls the flow of electricity, allowing the user to turn the circuit on or off. A **22k resistor (R1)** limits the current, protecting the transistor and the circuit from overloading. The **2N2222 transistor (Q1)** acts as a switch to amplify the current and drive the primary coil (L1). **L1 (with 3 turns)** and **L2 (with 275 turns)** form the coil system, where L1 induces a high-voltage signal in L2, creating the Tesla coil effect. The **LED** serves as an indicator, showing that the circuit is functioning correctly.

Nikola Tesla

Biography on Nikola Tesla

Nikola Tesla (1856-1943) was a pioneering inventor and electrical engineer known for his revolutionary contributions to electromagnetism and wireless communication. Born in what is now Croatia, Tesla emigrated to the United States and worked with Thomas Edison before establishing his own laboratory. He is best known for developing the alternating current (AC) system for electrical power distribution, which became the standard. Tesla also invented the Tesla coil, advanced radio technology, and experimented with wireless energy transmission. His visionary ideas laid the groundwork for modern electrical engineering and telecommunications, though he struggled with financial difficulties and died in relative obscurity.





THE END

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